

MAURITIUS EXAMINATIONS SYNDICATE

PSAC 2025

GRADE 6 MODULAR SCIENCE

Subject code: P141/2

Examiners' Report

INTRODUCTION

Until 2025, the assessment in Science was modular in nature with many candidates sitting for the first part of the assessment at the end of Grade 5 and the second part at the end of Grade 6. From 2026, candidates will sit for a single paper of 100 marks covering the content of both Grade 5 and Grade 6.

In addition to the syllabus content, candidates are assessed on three assessment objectives. Approximately 40% of the marks on the paper are for demonstrating knowledge and understanding of scientific ideas, scientific techniques and procedures (AO1), 40% on application of knowledge and understanding of scientific ideas (AO2) and approximately 20% for inquiry skills (AO3).

The purpose of the examiners' report for PSAC Grade 6 Science is to provide constructive feedback on candidates' performance and guidance to Educators and future candidates. The contents of the syllabus that gave rise to misconceptions are included in the report. The report also highlights gaps in the conceptual understanding of candidates. Other aspects which caused difficulty, along with possible reasons, are also commented on. Key messages are included to draw the attention of Educators on the essential areas to focus on.

This report should be read together with the question paper for the examination.

PAPER OVERVIEW

The 2025 Grade 6 Science paper consisted of six questions for a total of 50 marks. Question 1 comprised 5 multiple choice questions (MCQs), which covered the three assessment objectives. Questions 2-6 included objective-type questions as well as short-answer and open-ended questions that enable candidates to demonstrate their knowledge and application of essential scientific concepts.

KEY MESSAGES

- Language continues to be a barrier for some candidates, particularly in open-ended questions where they are required to explain, justify or describe scientific ideas clearly.
- Candidates should read questions carefully as they tend to lose marks due to misinterpretation or misreading of questions. In MCQ for example, some candidates selected more than one option, including the correct one. No mark is awarded in such cases when the choice of the answer is left to the examiner.
- Candidates need to avoid the telescoping of answers for questions requiring two separate responses.
- Although misspelling of scientific terms are often not penalised, candidates are encouraged to practise writing key scientific terms.

GENERAL COMMENTS

Most candidates showed good grasp of the syllabus content and there was a range of correct responses for the open-ended questions. They started the paper well but found the last items in each question more challenging. The last items are normally items pertaining to application questions and required candidates to apply the concepts learnt in an unknown or different situation that they have been used to. The application of concepts and questions assessing scientific inquiry skills remain a challenge for a majority of candidates at this level.

Items relating to AO1 were generally well answered. The weighting of the paper on the AOs ‘Application’ and ‘Inquiry skills’ is around 50%. It is, therefore, very important that these skills are developed along with basic knowledge with understanding.

The majority of candidates appeared to have sufficient time to complete the paper.

SPECIFIC COMMENTS

Question 1

Question 1 consists of 5 multiple choice questions and provided abroad coverage of the syllabus. **Table 1** provides the key and the most common incorrect choice for each item.

Table 1

<i>Item Number</i>	<i>Key</i>	<i>Most common incorrect choice</i>
1	B	A
2	D	A
3	C	D
4	B	C
5	A	C

This question was generally well attempted by most candidates. Items 1, 2 and 3 were the most accessible items while item 5 proved to be the most challenging one. Items 1, 2 and 3 are questions that required basic knowledge and understanding whereas the final item required more careful interpretation of the scientific conditions involved.

Comments on specific items

Item 1 was a relatively easy recall question on classification of animals. Candidates were required to identify a herbivore from a list of four animals. The correct answer was ‘cow’ and a good number of candidates selected the correct answer, B. This indicates that candidates are well acquainted with examples of animals that are herbivores. However, a common incorrect answer was birds.

Item 2 was a knowledge-based question. Candidates who identified D as the correct answer showed a good understanding of the solar system and the rotation of the Earth around the sun. A few candidates opted for ‘one day’, which was an incorrect answer. This could have arisen due to the fact that they pupils learn about the rotation of the Earth on its own axis takes 24 hours, that is a day.

Item 3 was an application question. Candidates were required to identify a balanced meal for a vegetarian from a list of four given pictures. Whilst the majority of the candidates could identify C as the correct answer, some chose D, which was an incorrect answer. The attention of candidates needs to be drawn to the fact that when words are highlighted as bold, this is to lay emphasis on key requirement of the question. Whilst, it seems that many candidates understood that there need to be three food groups for a balanced meal, they ignored that it was specifically for a vegetarian.

Item 4 was a simple application question on the topic ‘Animals’. Candidates were required to identify an animal that has three pairs of legs. The correct answer was B. Whilst a small number of candidates opted for mosquito, the majority opted for spider. It is important that students understand that spiders belong to the family of arachnids and that arachnids have 4 pairs of legs.

Item 5 was about the unit ‘Materials’. Options B and D proved to be clearly incorrect for the majority of candidates. This shows that the concept that rusting requires both water and air (oxygen) is quite clear to a good number of candidates. Many candidates chose C, which was incorrect. Candidates understood that rusting will take place in test tube Q. However, they failed to recognise that the iron nail in water with no air (oxygen) will not rust as there was a layer of oil that prevents the oxygen from air from dissolving in the water. The attention of pupils needs to be drawn to the fact that air consists of oxygen as well as water vapour, both conditions necessary for rusting.

Question 2

This question was mainly on the topic ‘Plants’. The first and second parts of the question were about photosynthesis. Candidates fared generally well on this question, with most of them scoring more than 5 out of 7 marks. This question was one of the most popular questions in the paper.

Comments on specific items

- (a) Many candidates could identify Gas A as carbon dioxide and Gas B as oxygen. This shows that the concept of gas exchange in plants has been well grasped.
- (b)(i) The question required candidates to give two other conditions necessary for photosynthesis apart from carbon dioxide. Many candidates could identify any two among sunlight, chlorophyll and water. A common mistake was mentioning carbon dioxide, revealing that enough attention was not paid while reading the question. Other common mistake was giving sun as an answer instead of sunlight.
- (b)(ii) Candidates were required to give the importance of photosynthesis to plants. A significant number of pupils gave the correct answer by mentioning that photosynthesis produces food for the plant. Due to misreading of the question, the weaker ability student ended up giving either the definition of photosynthesis or giving other importance of photosynthesis such as ‘photosynthesis helps to produce oxygen’.
- (b)(iii) This was a high-scoring item with the majority of candidates identifying respiration as the correct answer.
- (c) Candidates were required to identify the characteristic of fish that allows them to take in air dissolved in water. The answers revealed that candidates are well acquainted with the characteristics of fish as they could mention gills, fins and tails. However, the challenge was to identify the correct body part for the purpose mentioned in the question. The correct expected answer was the ‘gills’.

Question 3

This question was mainly on the topic ‘Materials’. Whilst the objective type question, the matching items, was well attempted by a majority of candidates., the second part on gold proved to be challenging.

Candidates fared generally well on this question, with most of them scoring more than 8 out of 12 marks.

- (a)(i)** This was generally well answered. This item was well attempted by the weaker ability candidates. The matching was well done with the expected answer being aluminium matched to Earth, Leather matched to animals and Wood matched to plants. One common misconception was that aluminium was man-made. Attention of pupils need to be drawn to the fact that although the options given on the right-hand side of the matching are more than that on the left-hand side, each item on the left should be matched to only one item on the right.
- (b)(i)** Most candidates answered this question well, showing that they could easily identify uses of gold. The most common answer was to make jewellery. Yet, it was observed that there were many spelling mistakes when writing the word ‘jewellery’. Some even gave answer in French/ Creole, which is to be discouraged.
- (b)(ii)** This item required candidates to state the property of gold that make it suitable for this use. The accepted answers were ‘Gold does not rust’, ‘Gold is rare’ or a ‘Gold is a malleable metal’ It is to be noted that many gave another use for this part rather than a property.
- (b)(iii)** Many candidates found it difficult to explain how gold can be made harder. The correct expected answer was to add other metal to gold to make it stronger. Common incorrect answers included melting the gold or transforming it into gold bars. A significant number of candidates left this question unanswered or answered the question only partly.
- (c)** Candidates were required to identify the materials used for windscreen and tyre of a car and to state a suitable property of each. The correct answer for windscreen was glass and for tyre it was rubber. There were many misspellings of the correct answers such as *Rudder*, *Ruber*, *Rubear*. Candidates were not penalised as long as the meaning was understandable and unchanged. However, answer such as *grass* instead of glass was not awarded any mark. Other common mistake was misplacing the property of glass for tyre and vice-versa. Candidates are encouraged to revise their answers to avoid losing marks unnecessarily.

- (d) Many candidates correctly identified that aluminium is light and does not rust. Low ability pupils did not understand the question. Candidates are encouraged to give one idea rather than two ideas in the same answer. In so doing, they tend to provide the correct answer and an incorrect answer in the same phrase/ sentence. In such cases, since the decision is on the examiner to choose the correct answer, marks are not awarded. A common incorrect answer is to state that aluminium is solid. It is true that aluminium is a solid metal, but that it is not the property that makes it suitable for making body parts of airplane.

Question 4

This question focussed mainly on the environment with the first part being on pollution, the second part on global warming and the last part of the question on the 3Rs.

- (a)(i) This item required candidates to identify two types of pollution from a diagram. Whilst many candidates correctly identified water pollution and air pollution, a recurring incorrect answer was sea pollution. Candidates should be encouraged to use the accepted scientific terms.
- (a)(ii) The majority of candidates could score the full marks on this item. The expected answers were smoke from the ship and wastewater from the factory. Some candidates, however, provided answers not visible from the diagram such as oil spill. The questions required the candidates to refer to the diagram. Candidates are also encouraged to provide detailed answer rather than vague one, for example smoke from the ship rather than the ship.
- (b) (i) This item required candidates to give the definition of global warming. It was observed that definition was challenging for many candidates. The expected answer was ‘an increase in average temperature of the Earth’s atmosphere’. While many candidates knew the meaning of global warming, they had difficulty in producing a correct answer. They missed an important part of the answer and ended up writing ‘an increase in the Earth atmosphere’, which unfortunately does not convey the idea of what is global warming.
- (b)(ii) Candidates were required to give two effects of global warming. There are many effects which were correct and were awarded marks. These include ‘rise in sea level’,

‘melting of icebergs’, ‘destruction of habitats’. However, an incorrect answer was to repeat the definition of global warming instead of giving its effects.

- (c)(i) This question was about good practices concerning waste disposal. Candidates had to identify a process that would convert used tin cans into useful products. The correct answer was ‘recycling’. Since the question was about identifying a process, reuse as an answer was not acceptable.
- (c)(ii) Candidates were required to give a use of manure. This proved to be challenging to them. The expected correct answer was to enrich the soil.
- (d) This item required candidates to provide a reason for not throwing used batteries in the environment. Candidates should be encouraged to provide accurate answers rather than general ones such as it causes pollution. They were expected to mention that the used batteries contain poisonous/ toxic substances, which are harmful to the environment.

Question 5

This question focussed on the unit ‘energy’ and included questions on forms of energy, renewable and non-renewable sources of energy and production of electricity in a thermal power station.

- (a) Candidates were required to identify the forms of energy at the input and output of a car running on petrol. The correct answer at the input was chemical energy and at the output, light/ sound/movement were correct answers. Many candidates opted for electrical energy at the input, which was not an acceptable answer. It was also observed that spelling mistakes for technical terms is recurrent with many candidates writing *Kemical* or *camical*. Candidates who did not pay careful attention when reading the stem wrote *petrol* or *fuel* as the form of energy at the input.
- (b) The question was about the advantage and disadvantage of using solar panels. The expected answer for the advantage was that solar energy is free or using solar panels does not cause pollution. Responding to open-ended questions and conveying their ideas correctly was a challenge for many candidates. Pupils should be advised to avoid ambiguous answers such as ‘it is free’ as there is no clear indication what does the

pronoun 'it' refers to. For the disadvantage part, the correct answer was that it is expensive to install.

- (c)(i) Candidates gave the definition of renewable source of energy instead of giving examples of renewable and non-renewable source of energy. Many candidates did not pay attention to the source of renewable fuel used in thermal power station and ended up giving incorrect answers such as sun, water and wind. The correct expected answer was bagasse for renewable source of energy. There were some candidates who mistook bagasse for a non-renewable source of energy. It needs to be brought to the attention of pupils that bagasse is a renewable source of energy but a polluting one.
- (c)(ii) This part of the question proved challenging to many candidates, both in terms of understanding the question and expressing themselves. The answers provided by the candidates revealed that while they knew the process, they had difficulty in identifying and reproducing the steps of the process in an orderly manner. In Step 2, the expected answer was 'production of steam' and Step 4, the expected answer was 'turns generator'. Candidates should be encouraged to provide at least a phrase for this type of question rather than one word answer such as steam and generator, which is not sufficiently indicative of what they want to convey.

Question 6

This question focussed on air and burning through an experimental situation involving candles covered by jars of different sizes.

- (a)(i) Many candidates scored their mark by mentioning oxygen. A common incorrect answer was fuel.
- (a)(ii) Most candidates correctly stated that carbon dioxide is produced during the burning of candles.
- (a)(iii) Candidates were required to identify the trend and make a prediction. This item assessed their inquiry skills. Acceptable answers generally were within the range of 1-3 seconds. Many candidates struggled to recognise the pattern and provided implausible answers.

- (a)(iv) Many candidates found it difficult to explain why all the candles stopped burning after some time. The expected answer was that all the oxygen was used up. There is a misconception about air and oxygen when it comes to burning. This results in many candidates mentioning that all air has been used up instead of oxygen being used up.
- (a)(v) The concept of control is poorly grasped by many candidates. There were very few good responses for this question. The expected answer was for the test to be a fair one.
- (b) Candidates were required to give two ways to extinguish fire. Whilst this was a simple recall question, candidates gave one-word answers such as sand and water which are not be to be encouraged. They were expected to write pouring water on the fire or using a fire extinguisher.

CONCLUSION AND RECOMMENDATIONS

This assessment suggests that there are some concepts that require some further reinforcement at Grade 6 level, particularly where candidates are asked to apply scientific knowledge. Experimentation forms an integral part in the teaching of science. Hands-on experiments and demonstrations are encouraged as they promote observation skills, which are primordial in the learning of Science. Experiments promote learning for all types of learners and linking the skills acquired during an experiment to various scenarios can make pupils relate scientific concepts to everyday life situations. It is definitely beneficial for pupils to carry out experiments in science laboratory/room. However, in the absence of a science room, it is encouraged that a science corner be created so that pupils have the opportunity to perform the experiments in their textbooks.

Open-ended questions remain a challenge for the majority of candidates. Pupils should be discouraged in class to give one-word answers for questions that require an elaborate answer. Language barrier is one major issue when it comes to these types of question. Pupils should be encouraged to read the questions attentively and extract the important information from the stem of the question. This will allow them to know exactly what is required from them in terms of answer.